

HackoWatt - Two Forecasting Architectures

One-page coding brief for the analytics team

Common target: Household Load = Base + Behaviour + Thermal | PV is modelled separately and applied after demand forecast

OPTION A - MODULAR / COMPONENT MODEL

Recommended primary architecture

Architecture

Base_t deterministic / simple cyclic baseline

Behaviour_t probabilistic appliances / occupancy driven

Thermal_t weather + heat-pump / physical model

Forecast = Base_hat + Behaviour_hat + Thermal_hat

Implementation rules

- Base: constant/cyclic loads; avoid ML unless needed.
- Behaviour: forecast expected appliance demand; random future events remain uncertain.
- Thermal: use predicted weather, set-point and house/heat-pump parameters.
- Keep component outputs so every peak can be explained by source.

Advantages

- High interpretability and direct peak explanation.
- Natural fit for cost optimisation and load shifting.
- Easy to stress-test weather, occupancy and appliance scenarios.

Main risk: more engineering and calibration effort.

OPTION B - HOLISTIC / DIRECT MODEL

Recommended benchmark / fallback

Architecture

Inputs X_t -> CatBoost / LightGBM / XGBoost ->

Total household load_hat(t)

Core feature set

- Calendar: hour, weekday, month, cyclical time features.
- History: lagged total load, rolling means/maxima.
- Weather forecast: temperature, irradiance/cloud variables if useful.
- Static scenario inputs: heating type, household/appliance configuration.

Advantages

- Fast to implement and usually strong on tabular forecasting.
- Single training pipeline and simple interface contract.
- Excellent benchmark against the modular architecture.

Limitations

Weaker causal/explanatory structure; harder to optimise individual loads.

Recommended implementation for the hackathon

Build both. Use Modular as the explainable primary system and Holistic as an independent benchmark.

1. Train / validate

Same time split for both. Report MAE, RMSE, 24h total error and peak-hour error.

2. Compare

Naïve baseline vs Modular vs Holistic. Evaluate separately for 1-24h, 25-72h and 73-168h.

3. Final output

Demand forecast + peak hour + reason. Separately forecast PV, then compute grid import/export and cost.